

0916 U1 sampling and surveys MCQ

1. A company that sells baby food is interested in the baby food preferences of all families with toddlers from a certain city. A representative from the company sets up a booth at one grocery store in the city that will be used to investigate baby food preference. The representative will ask people with toddlers who walk past the booth to complete the survey.

Which of the following best describes the sampling method the company will use?

- (A) Systematic random sampling
- (B) Simple random sampling
- (C) Stratified random sampling
- (D) Convenience sampling

**Answer D**

Correct. There is no random selection of people with toddlers. Instead, people with toddlers who walk by the booth in the grocery store are selected. This is a nonrandom selection that is convenient for the company to obtain, a nonrandom convenience sample.

2. In a certain school, students can choose whether to eat in the school's cafeteria. A reporter working for the school's newspaper polled students on their reactions to changes in the menu at the cafeteria. For each student leaving the cafeteria in one 30-minute time period, the reporter used a coin to determine whether to stop the student and ask how he or she felt about the new menu. In the reporter's article it was stated that a random sample of the students showed that 89 percent of the school's student population was happy with the new menu. Which of the following statements is true?
- (A) Because each student leaving the cafeteria was randomly selected and could choose to answer or not, this is a random sample of the student population, and the 89% is an accurate measurement of the school population's view of the new menu.
 - (B) Because students self-selected whether to eat in the cafeteria, the sampling method might be biased and the sample might not be representative of all students in the school.
 - (C) The survey would have been more effective if the reporter had collected the data in one 15-minute time period rather than in one 30-minute time period.
 - (D) The survey would have been more effective if students who cared about the food could have called the reporter to tell how they felt about the new menu, so that only students with opinions on the subject would have been surveyed.



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3. A large online retailer places packed and ready-to-be-shipped boxes in a line on a conveyor belt. A sample of boxes will be used to estimate the mean weight of boxes shipped on a given day. One proposed method for selecting the sample is as follows:
- Generate a random integer from 1 to 20.
 - Use that integer to select an initial box from the first 20 boxes in line on the conveyor belt.
 - Select every 25th box that follows the initial box on the conveyor belt for the rest of the day.

Which of the following best describes the proposed sampling method?

- (A) Stratified random sample, stratified by time of day
- (B) Cluster sample, with times of day as clusters
- (C) Cluster sample, with box weights as clusters
- (D) Systematic random sample

**Answer D**

Correct. A systematic random sample is a method in which sample members from a population are selected according to a random starting point and then a fixed, periodic interval.

4. A company is hired to identify the issues that all residents in a large district are concerned about. The company divides the district into twenty regions and believes that the regions are similar to one another in their composition of residents. The company then randomly selects two of the regions and surveys all residents in both regions.

Which of the following is the sampling method that will be used by the company?

- (A) Simple random sampling
- (B) Stratified random sampling
- (C) Systematic random sampling
- (D) Cluster random sampling

**Answer D**

Correct. A cluster random sample involves the division of a population into smaller groups, called clusters. Ideally, each cluster mirrors the heterogeneity of the population, with clusters similar to one another. A simple random sample of clusters is selected from the population to form the sample of clusters. Data are collected from all observational units in each of the selected clusters.

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5. A total of 13 distinct groups, but with each group similar to the other groups, participated in a craft fair. To investigate how satisfied the participants were with the fair, the fair director conducted a survey. The director randomly selected 5 of the groups participating in the fair and surveyed all members of the selected groups.

Which statement best describes the sampling method used by the director?

- (A) It was convenience sampling, because participants from each group were easy to contact.
- (B) It was simple random sampling, because the participants were selected at random.
- (C) It was stratified random sampling, because the director ensured that members from each group were selected.
- (D) It was cluster sampling, because the sample consisted of all members of the 5 randomly selected groups. ✓

Answer D

Correct. The sampling method is cluster sampling. The 13 distinct groups are clusters of craft fair participants similar to one another in their composition. A simple random sample of 5 groups (clusters) from the population of 13 groups is selected to form the sample of groups. Data are collected from all participants in the selected groups.

6. A researcher conducting a telephone survey is concerned about possible sources of bias. Of the following, which is the best example of nonresponse bias?
- (A) The wording of the questions in the survey leads people to respond in a certain way.
 - (B) People might be uncomfortable with the survey questions and, as a result, might not always respond to those questions truthfully.
 - (C) Many of the people selected to participate in the survey who do not respond might have opinions different from those who do respond. ✓
 - (D) People without telephones are overlooked in the sampling procedure used to determine who is surveyed.
7. In the design of a survey, which of the following best explains how to minimize response bias?
- (A) Increase the sample size.
 - (B) Decrease the sample size.
 - (C) Randomly select the sample.
 - (D) Carefully word and field-test survey questions. ✓

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8. Under which of the following conditions is it preferable to use stratified random sampling rather than simple random sampling?
- (A) The population can be divided into a small number of strata so that each stratum contains a large number of individuals.
 - (B) The population can be divided into strata so that the individuals in each stratum are as much alike as possible. ✓
 - (C) The population can be divided into strata so that the individuals in each stratum are as different as possible.
 - (D) The population can be divided into strata of equal sizes so that each individual in the population still has the same chance of being selected.
9. A certain county school district has 15 high schools. The high school seniors' plans after graduation in each school vary greatly from one school to the next. The county superintendent will select a sample of high school seniors from the district to survey about their plans after graduation. The superintendent will use a cluster sample with the high schools as clusters. A random sample of 5 high schools will be selected, and all seniors at those high schools will complete the survey. What is one disadvantage to selecting a cluster sample to investigate the superintendent's goal?
- (A) Because every senior in the selected clusters will complete the survey, the sample will be too large to yield accurate results.
 - (B) The schools in the cluster sample might not be representative of the population of seniors. ✓
 - (C) There could be seniors absent from school on the day the survey is given, which could affect the results.
 - (D) There is no disadvantage to using a cluster sample.

Answer B

Correct. Because of the known difference between schools in terms of plans after graduation, the schools not chosen may have seniors that differ in some way from the seniors in schools that are chosen. Therefore, the plans of the students surveyed may not accurately represent the population in the district.

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10. To estimate the percent of red marbles in a large bag of marbles, Margo will use the following sampling method. She will randomly select a marble, record its color, put it back into the bag, shake the bag to thoroughly mix the marbles, and then repeat those steps. She will perform the procedure many times. What type of sampling method is Margo using?
- (A) Stratified random sampling
 - (B) Systematic random sampling
 - (C) Random sampling with replacement ✓
 - (D) Random sampling without replacement

Answer C

Correct. Each marble selected is randomly selected, so the sampling is random. Since the chosen marble is placed back into the bag before the next selection of a marble, the sampling is done with replacement. Margo is using random sampling with replacement.

11. Each person in a simple random sample of 2,000 received a survey, and 317 people returned their survey. How could nonresponse cause the results of the survey to be biased?
- (A) Those who did not respond reduced the sample size, and small samples have more bias than large samples.
 - (B) Those who did not respond caused a violation of the assumption of independence.
 - (C) Those who did not respond were indistinguishable from those who did not receive the survey.
 - (D) Those who did respond may differ in some important way from those who did not respond. ✓
12. At a certain clothing store, the clothes are displayed on racks. The clothes on each rack have similar prices, but the prices among the racks are very different. To estimate the typical price of a single piece of clothing, a consumer will randomly select four pieces of clothing from each rack. What type of sample is the consumer selecting?
- (A) A census
 - (B) A simple random sample
 - (C) A stratified random sample ✓
 - (D) A systematic random sample

Answer C

Correct. The pieces of clothing in the store are divided into strata, where each stratum is a rack of clothes having similar prices. The mean price calculated using the randomly selected pieces of clothing from the strata is an example of using a stratified random sample.

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13. In a recent poll of 1,500 randomly selected eligible voters, only 525 (35 percent) said that they did not vote in the last election. However, a vote count showed that 80 percent of eligible voters actually did not vote in the last election. Which of the following types of bias is most likely to have occurred in the poll?
- (A) Nonresponse bias
 - (B) Sampling bias
 - (C) Response bias ✓
 - (D) Undercoverage bias
14. A school principal wanted to investigate student opinion about the food served in the school cafeteria. The principal selected at random samples of 50 first-year students, 50 second-year students, 50 third-year students, and 50 fourth-year students to complete a questionnaire. Which of the following best describes the principal's sampling plan?
- (A) A stratified random sample ✓
 - (B) A simple random sample
 - (C) A cluster sample
 - (D) A systematic sample
15. A newspaper editor wants to investigate whether residents of the city support a proposal to build a new high school football stadium. The editor hires a polling firm to conduct a survey and requests that a sample of 500 residents be selected using a stratified sampling design based on voting districts within the city. Which of the following methods will achieve the desired sampling design?
- (A) Send a survey to all city residents and use the first 500 returned surveys for the sample.
 - (B) Select a random sample from each voting district based on the proportion of city residents in the district so that a total of 500 is obtained. ✓
 - (C) Select one voting district at random, and then select a random sample of 500 from the selected voting district.
 - (D) Alphabetize a list of all city residents, and then select the first 500 residents on the list, classifying those selected by voting district.

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16. Which of the following does not describe a sampling method that has a potential source of voluntary response bias for the administration of a survey about college athletics at a university?

- (A) Giving the survey to 30 students selected at random from each of the eight dorms on campus ✓
- (B) Giving the survey to the first 50 students who call the university radio station after the broadcaster announces the survey on air
- (C) Giving the survey to any student who stops at a table set up in the student union after the survey is advertised
- (D) Giving the survey to physical education majors who receive extra credit for completing the survey

Answer A

Correct. Since the 30 students are selected at random, they are not volunteers, so the selection of those students is not a likely source of voluntary response bias.

17. The student government at a high school wants to conduct a survey of student opinion. It wants to begin with a simple random sample of 60 students. Which of the following survey methods will produce a simple random sample?

- (A) Survey the first 60 students to arrive at school in the morning.
- (B) Use random numbers to choose 15 each of first-year, second-year, third-year, and fourth-year students.
- (C) Number the cafeteria seats. Use a table of random numbers to choose seats and interview the students until 60 have been interviewed.
- (D) Number the students in the official school roster. Use a table of random numbers to choose 60 students from this roster for the survey. ✓

18. An independent polling agency was hired to track the preferences of registered voters in a district for an upcoming election. The polling agency divided the district into twenty regions and believes that the regions are similar to one another in their composition. The agency then randomly selected two of the regions and surveyed all registered voters in both regions.

Which of the following best describes the sampling method used by the polling agency?

- (A) Simple random sampling
- (B) Stratified random sampling
- (C) Systematic sampling
- (D) Cluster Sampling ✓

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19. A school nutritionist was interested in how students at a certain school would feel after taking a nutritional supplement. The nutritionist selected a random sample of twenty students from the school to participate in the study. Participants were asked to keep a journal on how well they felt after taking the supplement each day. What possible source of bias is present in the method of data collection?
- (A) Undercoverage bias
 - (B) Nonresponse bias
 - (C) Response bias where responses are self-reported ✓
 - (D) Response bias where the question wording is leading or confusing

Answer C

Correct. The participants were asked to assess and record how they felt, making it more likely they could overestimate or underestimate the effects of the supplement.

20. As part of a demographic study, a college administrator needed to survey a sample of students from the college. From each major offered at the college, the administrator randomly selected 5 percent of the students with that major to participate in the survey.
- Which of the following is the best description of the type of sample selected by the administrator?
- (A) Cluster sample
 - (B) Simple random sample
 - (C) Stratified random sample ✓
 - (D) Systematic random sample

Answer C

Correct. The administrator selected a stratified random sample, because all of the students at the college were separated into strata (the majors) and a random sample was selected from each of the strata.